MecHand — drawing reaction mechanisms by hand

A tutorial

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Abstract

MecHand lets you draw reaction mechanisms as you would on paper — structures, charges and curved electron-flow arrows — with a pen, a trackpad or a mouse. It recognizes the drawing in your web browser, checks the electron flow, places the result in the Ketcher chemical editor, and exports it to LaTeX (TikZ and chemfig), ChemDraw and Marvin. This tutorial takes you from opening MecHand to a mechanism figure in your own LaTeX document in about fifteen minutes.

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1 Opening MecHand

Open `compbiochbiophlab.github.io/MecHand/app` in a current version of Chrome, Edge, Firefox or Safari, on a computer or a tablet. Nothing is installed: the chemical editor, its chemistry engine and the handwriting recognizer all run inside the browser, and your drawings never leave your computer. The first visit downloads about 9 MB, which the browser then keeps.

The MecHand bar (Figure 1) has these controls:

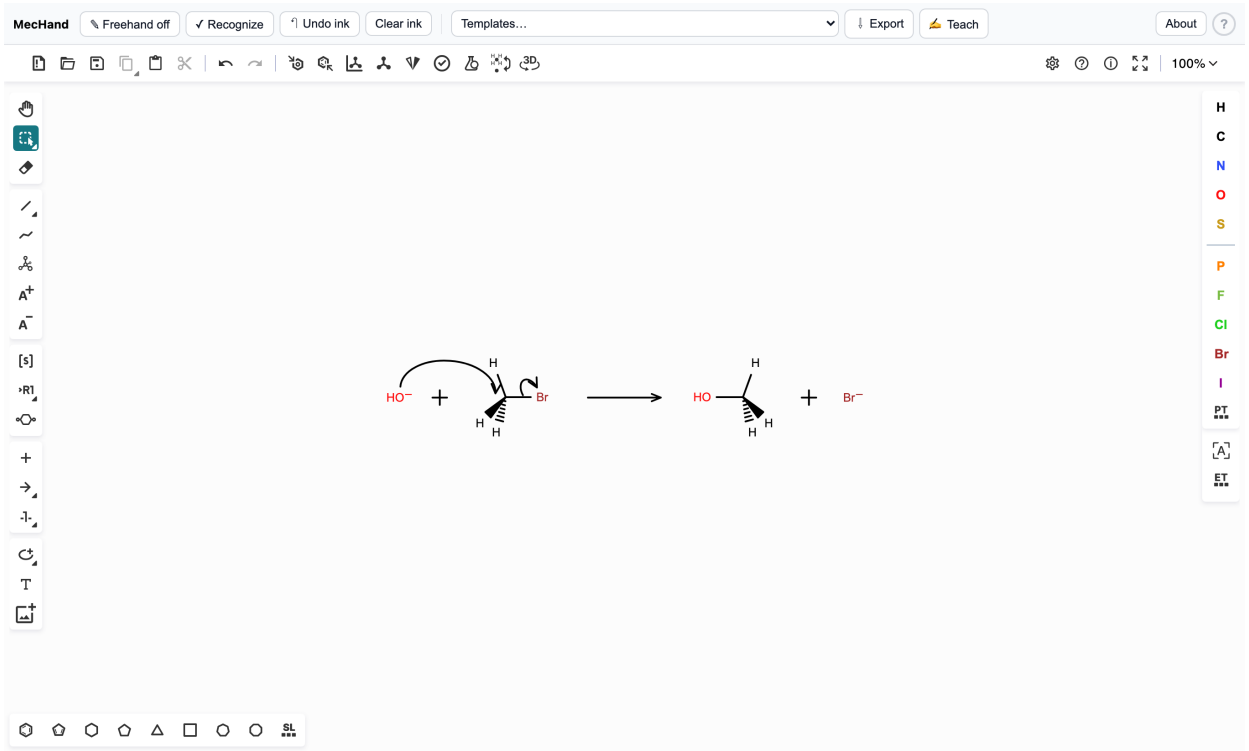


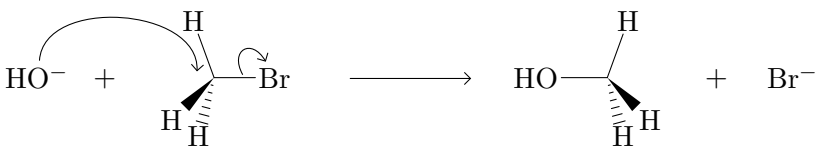
Figure 1: MecHand with the S_N2 template open. The MecHand bar is at the top; below it is the Ketcher editor with its own toolbars.

Control	Purpose
Freehand `Alt/Option F`	Turns the pen layer on and off. When it is off, Ketcher's tools work as usual.
Recognize `Space`	Reads your ink and shows what MecHand understood.
Undo ink, **Clear ink**	Remove the last stroke, or all strokes not yet inserted.
Templates…	Opens one of 13 standard mechanisms.
Export	TikZ, chemfig, ChemDraw (CDXML), MOL/RXN, KET or SMILES.
Teach	Teaches MecHand your handwriting.
About, **?**	Recognition details and browser notes; help labels on every button.

Running it on your own computer (optional). Install Node.js (on macOS: `brew install node`; elsewhere, the installer from `https://nodejs.org`), then run `npm ci` and `npm run dev` in the `ketcher-mechhand` folder of the repository and open `http://localhost:5173`.

2 Your first mechanism: S_N2

We will draw the reactants of the S_N2 reaction of hydroxide with bromomethane, with its two curved arrows:



1. Press **Freehand** (or `Alt/Option F`). The canvas turns faintly blue: you are drawing ink.
2. Write **O** with a small − up and to its right: hydroxide.

3. Write a +, then draw the carbon: three short lines ending at three **H** and one line ending at **Br**. Leave a small gap between each line and its letter, as in a printed formula, and make the lines about two or three times longer than the letters are tall.

4. Draw the first curved arrow from the oxygen to the carbon, and finish it with a *full* arrowhead — a small "V", drawn without lifting the pen or as a separate stroke.

5. Draw the second curved arrow from the middle of the C–Br bond to the bromine.

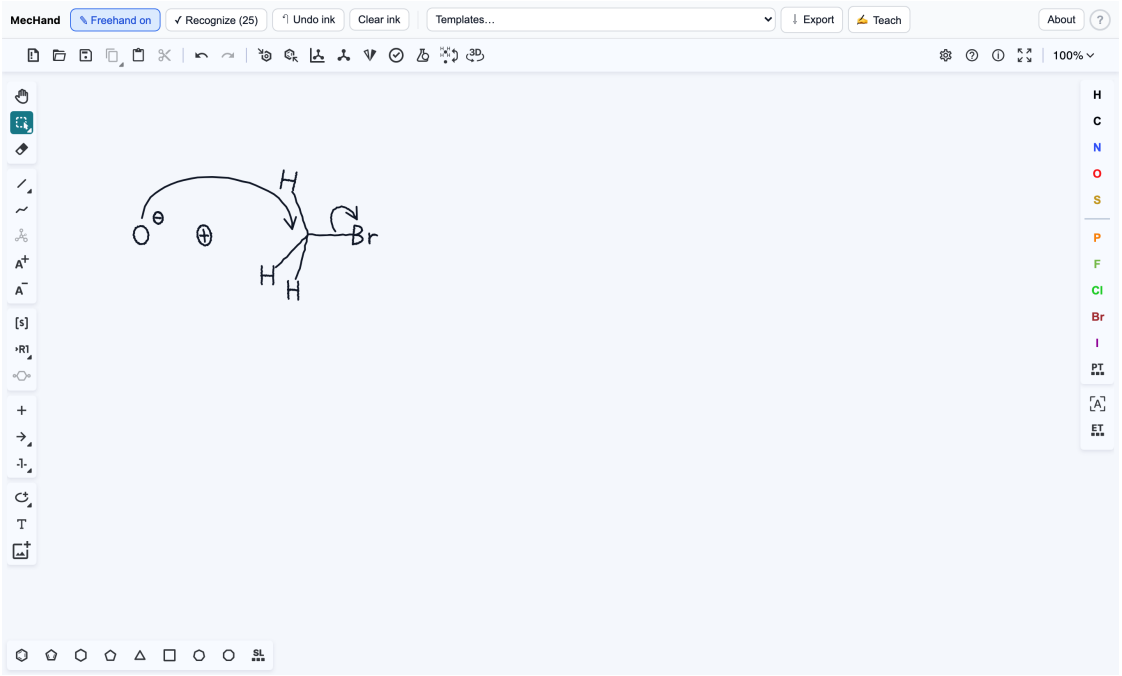


Figure 2: The S_N2 reactants drawn as ink.

Now press **Recognize** (or `Space`).

3 Is this what you meant?

MecHand colours your ink by how it read each stroke — blue bonds, green labels, red arrows, grey not understood — and shows a clean drawing of the result with its confidence (Figure 3).

Insert into Ketcher adds the structure and the arrows to the canvas. They are now ordinary Ketcher objects: move them, add stereochemistry, or keep drawing with Ketcher's tools.

Keep drawing closes the dialog and keeps the ink, so you can fix or add strokes.

Discard ink deletes the strokes.

Electron-flow check. Each curved arrow is attached to the atom or bond nearest to its ends. The dialog warns you when an electron-pair arrow starts at an atom with no lone pair (start it at a lone pair or at a bond instead), when an arrow starts or ends on nothing, or when an arrow would give C, N, O or F more than an octet while no bond to that atom breaks in the same step. Try it: draw an arrow from the CH_3 carbon to the oxygen and recognize again.

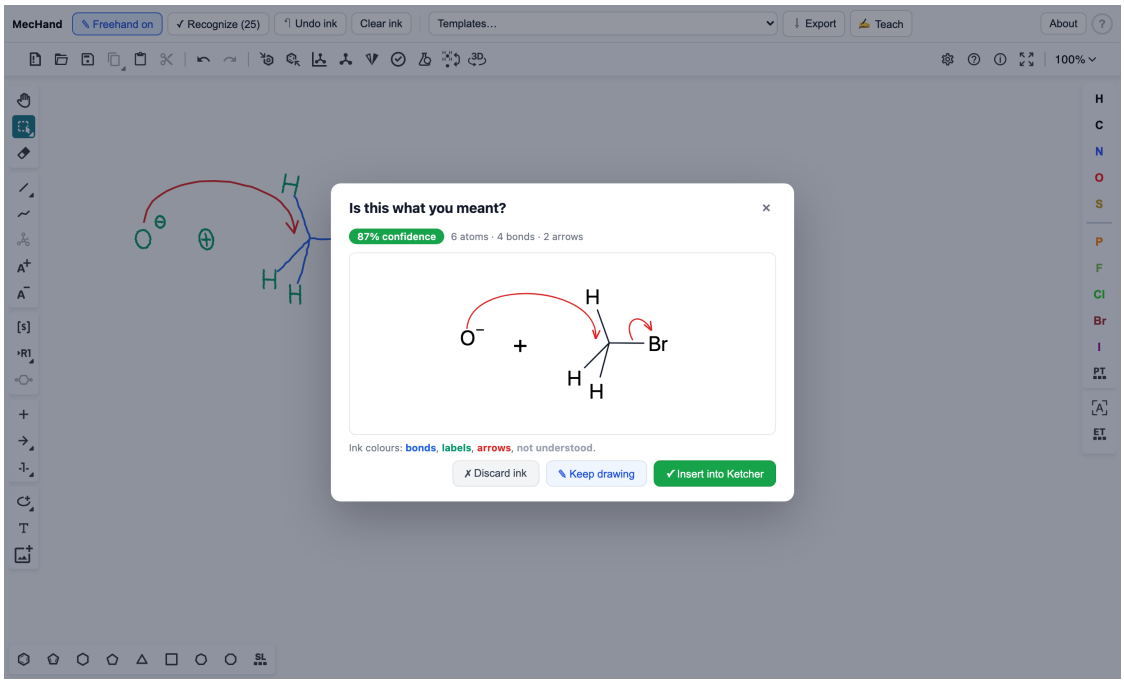


Figure 3: The recognition dialog: 6 atoms, 4 bonds and 2 arrows, with 87 % confidence.

What MecHand understands

You draw	MecHand reads
a line; a zig-zag or polygon in one stroke	bonds; a chain or a ring
two or three close parallel lines	a double or triple bond
letters	atom labels: C, H, N, O, S, P, F, Cl, Br, I, B, Si, Li, Na, K, Mg, Al, Fe, Cu, Zn, Pt, Pd…
+, −, ⊕, ⊖, 2−, 3+ at the upper right of an atom	a formal charge
a single dot next to an atom	a radical
a + between molecules	a reaction plus sign
a curve with a full head / a single barb	an electron-pair / a single-electron arrow
a long straight arrow	a reaction arrow

4 Templates: practise with standard mechanisms

Templates… opens 13 textbook mechanisms: S_N2, E2, addition to a carbonyl, aromatic nitration, Diels–Alder and Cl_2 homolysis (organic); the NH_3–BF_3 Lewis adduct, associative substitution at Pt(II), hydrolysis of $SiCl_4$ and the acidity of $[Fe(OH_2)_6]^{3+}$ (inorganic); and serine-protease acylation, alcohol dehydrogenase and a kinase (biochemistry). A good exercise: open a template, study it, clear the canvas, and redraw it by hand. The Diels–Alder template, as exported to TikZ:



5 Exporting to LaTeX

Export converts everything on the canvas — including curved arrows — into code (Figure 4). **TikZ** gives absolute coordinates and Bézier arrows that are easy to adjust; **chemfig** gives a reaction scheme with `\chemmove` arrows. Copy the code, or download it, and paste it into your document together with the preamble lines shown at the top:

```
\usepackage{tikz}
\usetikzlibrary{arrows.meta,bending}
\usepackage{chemfig}                    % for the chemfig export
\usetikzlibrary{calc}                   % (chemfig arrows)
```

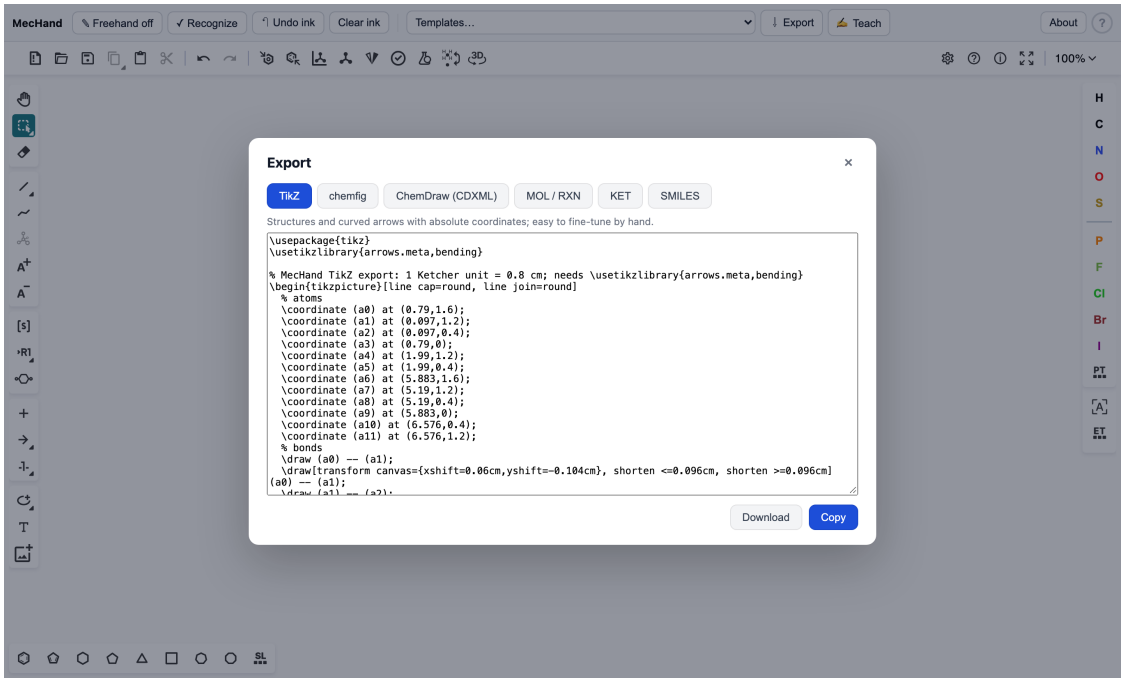
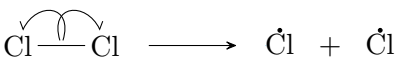


Figure 4: The Export dialog with the TikZ code of the Diels–Alder template.

The figures in this tutorial were produced exactly this way. Here is the homolysis of Cl_2 exported to chemfig (compile twice: chemfig places the arrows using positions recorded in the first run):



The other formats — CDXML, MOL/RXN, KET and SMILES — open in ChemDraw, Marvin, RDKit and other software. Ketcher also opens CDXML, MOL, RXN and SMILES files, so a mechanism started elsewhere can be finished in MecHand.

6 Teaching MecHand your handwriting

If a letter is often misread, open **Teach**, choose the symbol, write it in the box and press **Save sample**; a few samples per symbol help. Samples are stored in your browser and used at once. **Export samples** saves them as a file that can be added to MecHand's shared model, so the recognizer improves for everybody (see the website, "Handwriting recognition").

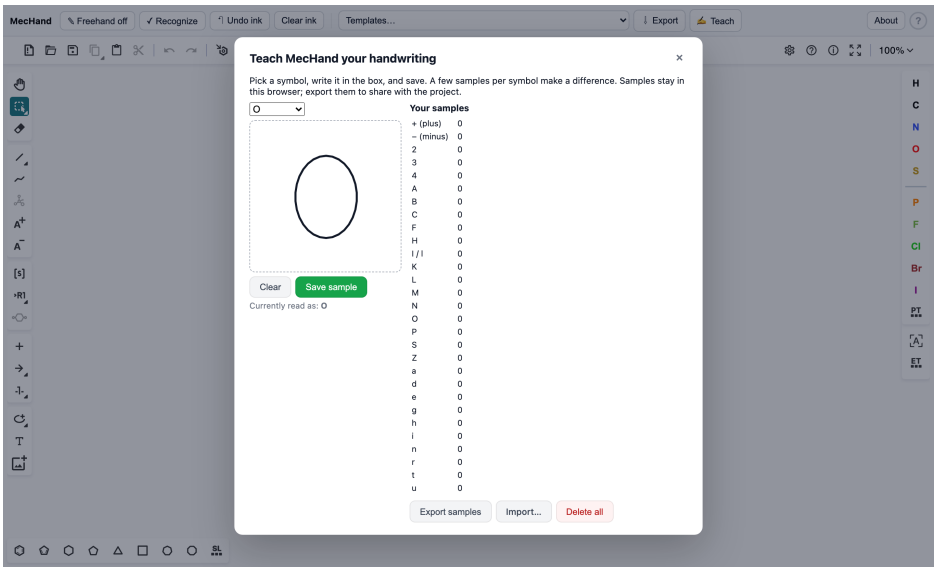


Figure 5: Teach: the box shows how your sample is currently read.

7 Tips and troubleshooting

- **Bonds read as letters, or letters as bonds?** Draw bonds longer than letters are tall, and leave a small gap between a bond and its label.
- **An arrow is not recognized?** Make the head clearly visible: a V whose arms point back along the shaft. A single barb means a one-electron (fishhook) arrow.
- **Panning and zooming** the canvas with touch or pen only works with freehand mode off.
- **Undo:** in freehand mode `Ctrl/Cmd Z` removes the last stroke; otherwise it is Ketcher's undo.
- **Browsers:** current Chrome, Edge, Firefox and Safari, on computers and tablets (with Apple Pencil, S Pen and other styluses). Safari and iPadOS before 18.2 record fewer points per stroke, which makes fast ink look more angular but does not stop recognition.
- **Optional image-based recognition:** MecHand can also send your ink to DECIMER, a deep-learning model, running on your own computer (**About** → Structure engine). See "Optional DECIMER server" on the website.

Website and documentation: `https://compbiochbiophlab.github.io/MecHand/` · Source code: `https://github.com/CompBiochBiophLab/MecHand` · MecHand is free software (MIT); Ketcher is © EPAM Systems (Apache 2.0).